Gojek Phone Masking: Telephony Architecture

Virtual Number Pooling, Bipartite Graph Allocation & Sub-100ms Telco Routing

PRIVACY ENGINEERING
TELEPHONY SCALING

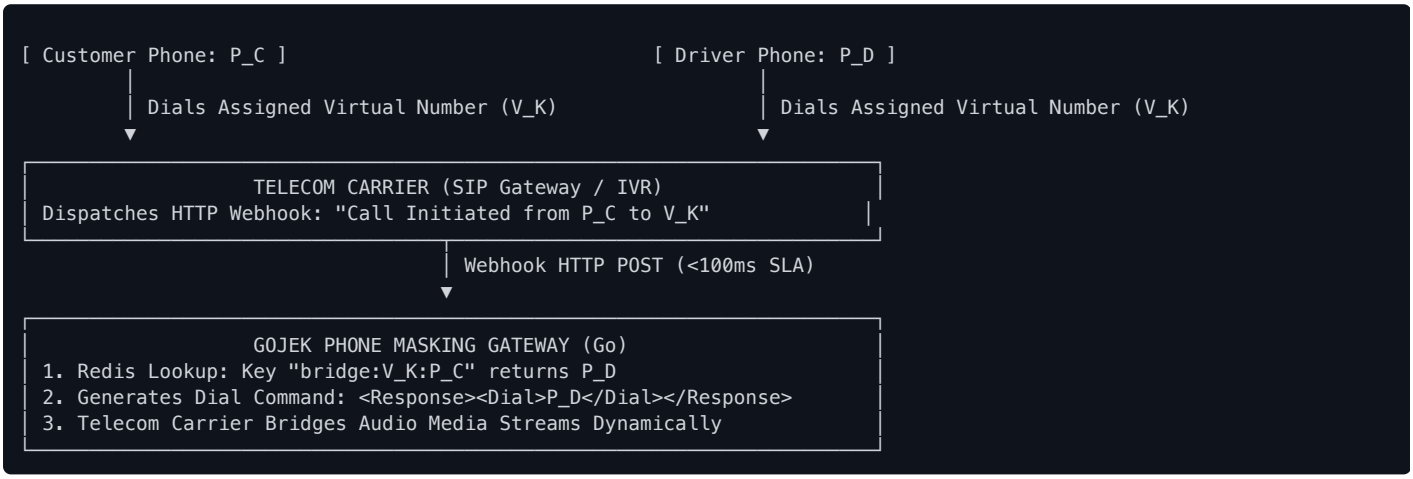
1. The Privacy Problem Space in On-Demand Logistics

Gojek processes millions of rides, food orders, and parcel deliveries daily. Drivers and consumers frequently need to communicate to coordinate pickups and delivery locations.

The Danger of Exposing Personal Mobile Numbers
Displaying raw phone numbers in driver or customer apps exposes users to stalking, harassment, scam campaigns, and data privacy regulatory penalties. Gojek enforces number masking to bridge audio calls through anonymous virtual numbers while keeping personal numbers strictly concealed.

2. Virtual Bridge Architecture & Telecom Interfacing

Gojek acts as an intelligent routing brain between telecommunication carriers (Twilio, Exotel, local SIP trunks) and user devices:



3. The Number Pooling Challenge: Millions of Rides vs. Few Numbers

Telecom providers charge monthly recurring lease fees for virtual DID numbers. Maintaining millions of dedicated numbers is economically impossible:

- **The Multiplexing Principle:** A single virtual number V_K can be simultaneously assigned to Order 1 (Customer A & Driver B) and Order 2 (Customer C & Driver D), because if Driver B dials V_K, the caller's Caller Line Identification (CLI) identifies Order 1 without ambiguity.

4. Graph Coloring & Conflict-Free Allocation Algorithm

Virtual number assignment is modeled as a vertex coloring problem on an active trip graph:

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Conflict Condition:
Two concurrent orders O_1(Customer_A, Driver_B) and O_2(Customer_C, Driver_D) CANNOT
share virtual number V_K if:
                    {Customer_A, Driver_B} ∩ {Customer_C, Driver_D} ≠ ∅

LRU Recycling Protocol:
1. Available virtual numbers reside in a Redis Sorted Set scored by last order end-time.
2. The allocator picks the number with the oldest release timestamp (longest cooldown).
3. Guarantees lagging callbacks from previous trips do not misconnect to new parties.
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5. Sub-100ms In-Memory Key Schema in Redis

Telecom switches enforce strict 1,500ms timeouts; if an IVR webhook does not answer, the switch terminates the call. Gojek achieves sub-100ms lookups using symmetric Redis keys:

Bidirectional Redis Mapping

`bridge:{VirtualNum}:{CustomerPhone}` ⟶ `DriverPhone`
`bridge:{VirtualNum}:{DriverPhone}` ⟶ `CustomerPhone`
Enables $O(1)$ constant-time lookup regardless of who initiates the call.

Adaptive Lifecycle TTL

Initialized with a 2-hour TTL during the trip. Upon trip completion, TTL is shortened to 15 minutes, preserving a brief callback window for lost items before recycling.

6. Architectural Comparison: Number Allocation Models

Strategy	Number Pool Size Needed	Monthly Cost	Collision Risk
Dedicated 1:1 Numbers	Equal to active orders (Millions)	Prohibitive ($ millions/mo)	Zero
Gojek Dynamic Multiplexing	**1,000 – 5,000 Numbers**	**Ultra-Low (< 0.1% cost)**	**Zero (CLI-disambiguated)**
Static Random Pooling	10,000 Numbers	Moderate	High (Cross-talk on multi-orders)

7. Telephony Failover & Fallback Protocols

Telecom carrier trunks are notorious for localized fiber cuts and regional routing outages:

- **Multi-Carrier SIP Trunk Redundancy:** The gateway monitors carrier webhook response times. If Carrier A experiences elevated error rates, inbound traffic fails over to Carrier B within 5 seconds.
- **In-App VoIP Fallback (WebRTC):** If PSTN cellular networks fail, mobile clients automatically offer an encrypted in-app voice call via WebRTC, bypassing telecom carriers entirely.
- **Automated Call Recording & Sentiment NLP:** Telephony audio streams are recorded and analyzed for abusive sentiment, flagging harassment to trust & safety agents within minutes.

8. Key Architectural Takeaways for Telephony Systems

Core Tenets Distilled from Gojek Phone Masking

- **Multiplex Scarce External Resources:** Use combined identifiers (Caller CLI + Virtual DID) to allow a tiny pool of expensive virtual numbers to service millions of simultaneous orders.
- **Telecom Gateways Require Sub-100ms Latency:** Telephony switches disconnect silent lines after 1.5 seconds. Use in-memory key-value stores (Redis) and lightweight Go handlers to respond within milliseconds.
- **Enforce Strict Cooldown Windows:** Reclaiming virtual numbers too quickly causes confusing callbacks from past customers to reach new drivers. Always implement an LRU cooldown buffer.
- **Bridge the Gap with VoIP Fallbacks:** Cellular PSTN lines are subject to carrier outages. In-app WebRTC voice channels guarantee high availability regardless of telecom provider reliability.